

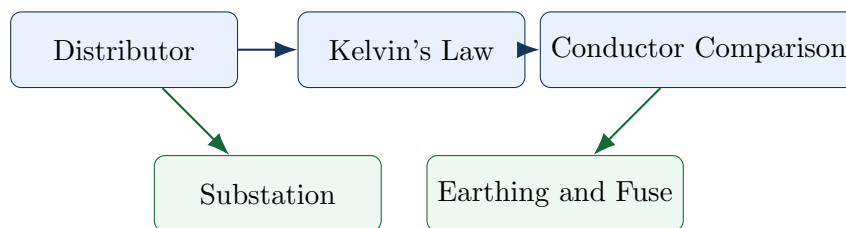
B.E. Electrical Engineering

Power Supply Systems

2023 Part-II Complete Solutions

Academic Article Style Notes

Simple English, SN ma'am approach, neat labelled diagrams, derivations, and stepwise numericals.



Prepared for semester-answer writing
May 16, 2026

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How to Read This Solution Set

Answer-writing rule

For Part-II of Power Supply Systems, write answers in this order:

Simple definition → neat diagram → assumptions → formula → stepwise calculation → final answer

This follows the same pattern used in SN ma'am notes.

Questions covered

This document solves every Part-II question from the 2023 Power Supply Systems paper:

- Group A: Questions 1 to 5, including all sub-parts.
- Group B: Questions 6 to 8, including all sub-parts.

1 Group A, Question 1(i): Minimum Potential Point in a Uniformly Loaded Distributor with Unequal End Voltages

Question 1(i)

Derive an expression for the distance of the point of minimum potential in a uniformly loaded distributor fed at both ends with unequal voltages.

Basic idea

In a distributor fed from both ends, current enters from both sides. The voltage decreases from one end, reaches a minimum value, and then increases towards the other end. The point where current changes direction is called the **point of minimum potential**.

Neat labelled diagram

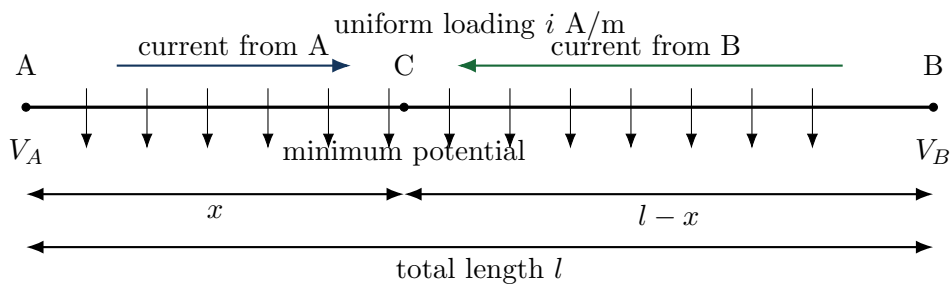


Figure 1: Uniformly loaded distributor fed at both ends with unequal voltages.

Assumptions

- Total length of the distributor is l metre.
- Uniform loading is i ampere per metre.
- Resistance of distributor per metre, including go and return path, is r ohm per metre.
- Voltage at feeding end A is V_A .
- Voltage at feeding end B is V_B .
- Point C is the point of minimum potential at distance x from A.

Derivation

At C, the current changes direction. Therefore, C is supplied partly from A and partly from B. Current supplied from A up to C:

$$I_A = ix$$

Current supplied from B up to C:

$$I_B = i(l - x)$$

Voltage drop from A to C for uniformly distributed load is

$$\Delta V_{AC} = \frac{irx^2}{2}$$

Voltage drop from B to C is

$$\Delta V_{BC} = \frac{ir(l - x)^2}{2}$$

Voltage at point C from A side is

$$V_C = V_A - \frac{irx^2}{2}$$

Voltage at point C from B side is

$$V_C = V_B - \frac{ir(l - x)^2}{2}$$

Equating both expressions of V_C ,

$$V_A - \frac{irx^2}{2} = V_B - \frac{ir(l - x)^2}{2}$$

$$V_A - V_B = \frac{ir}{2} [x^2 - (l - x)^2]$$

Now,

$$x^2 - (l - x)^2 = x^2 - (l^2 - 2lx + x^2) = 2lx - l^2$$

So,

$$V_A - V_B = \frac{ir}{2}(2lx - l^2)$$

$$V_A - V_B = ir l \left(x - \frac{l}{2} \right)$$

$$\boxed{x = \frac{V_A - V_B}{ir l} + \frac{l}{2}}$$

Distance of minimum potential point from A is $x = \frac{V_A - V_B}{ir l} + \frac{l}{2}$.

Exam remark

If $V_A = V_B$, then $x = l/2$. So, for equal end voltages, the minimum potential point is exactly at the middle of a uniformly loaded distributor.

2 Group A, Question 1(ii): DC Two-Wire System Converted to Three-Phase Three-Wire AC System

Question 1(ii)

A d.c. two-wire system is to be converted to an a.c. three-phase three-wire system by addition of a third conductor of the same size as the two existing conductors. Determine the percentage additional load which the three-phase conductors can carry at 0.9 power factor lagging, for the same voltage between conductors and the same percentage power loss.

Core concept

The same percentage loss means:

$$\frac{\text{line loss}}{\text{power transmitted}} \text{ is same in both systems.}$$

Here the conductor resistance of each wire is same because the third conductor is similar to the existing two conductors.

Simple comparison diagram

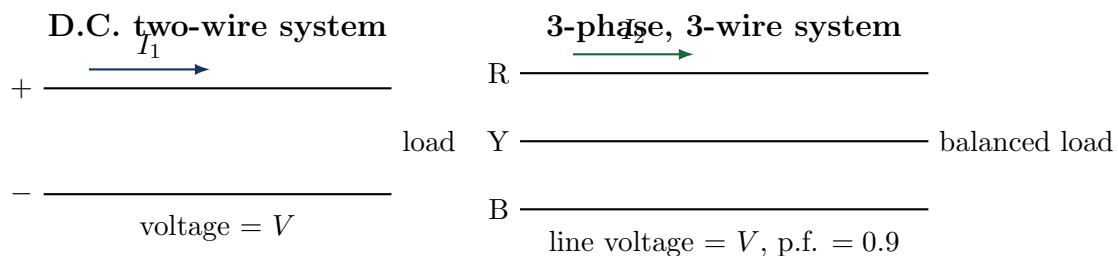


Figure 2: Conversion from d.c. two-wire to three-phase three-wire system.

Let

R = resistance of each conductor, V = voltage between conductors.

Let current in the d.c. system be I_1 and line current in the three-phase system be I_2 .

For d.c. two-wire system

Power transmitted:

$$P_1 = VI_1$$

Line loss:

$$W_1 = 2I_1^2R$$

Percentage loss is proportional to

$$\frac{W_1}{P_1} = \frac{2I_1^2 R}{VI_1} = \frac{2I_1 R}{V}$$

For three-phase, three-wire a.c. system

Power transmitted:

$$P_2 = \sqrt{3}VI_2 \cos \phi$$

Line loss:

$$W_2 = 3I_2^2 R$$

Percentage loss is proportional to

$$\frac{W_2}{P_2} = \frac{3I_2^2 R}{\sqrt{3}VI_2 \cos \phi} = \frac{\sqrt{3}I_2 R}{V \cos \phi}$$

Same percentage power loss

$$\frac{2I_1 R}{V} = \frac{\sqrt{3}I_2 R}{V \cos \phi}$$

$$I_2 = \frac{2I_1 \cos \phi}{\sqrt{3}}$$

Now,

$$\frac{P_2}{P_1} = \frac{\sqrt{3}VI_2 \cos \phi}{VI_1}$$

Substituting I_2 ,

$$\frac{P_2}{P_1} = \frac{\sqrt{3} \left(\frac{2I_1 \cos \phi}{\sqrt{3}} \right) \cos \phi}{I_1}$$

$$\frac{P_2}{P_1} = 2 \cos^2 \phi$$

Given,

$$\cos \phi = 0.9$$

$$\frac{P_2}{P_1} = 2(0.9)^2 = 1.62$$

Thus the new load is 1.62 times the old load.

Percentage additional load:

$$= (1.62 - 1) \times 100 = 62\%$$

The three-phase system can carry 62% additional load.

3 Group A, Question 2: Conductor Material Comparison for D.C. Two-Wire and Two-Phase Three-Wire Overhead Systems

Question 2

Compare the volume of conductor material required for d.c. two-wire system with one conductor earthed and two-phase three-wire system in overhead lines. State assumptions.

Assumptions

1. Same power P is transmitted in both systems.
2. Same length l of line is used.
3. Same line loss P_{loss} is allowed.
4. Same material of conductor is used, so resistivity ρ is same.
5. In overhead systems, maximum voltage between any conductor and earth is kept same and equal to V_m .
6. For a.c. system, power factor is $\cos \phi$.

Case 1: D.C. two-wire system with one conductor earthed

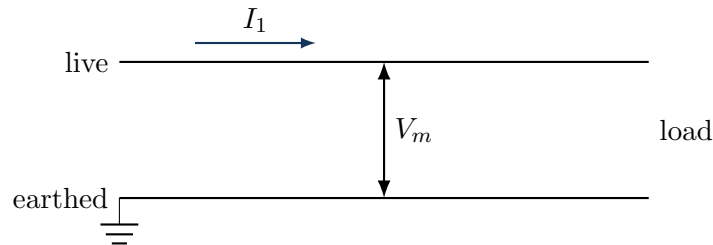


Figure 3: D.C. two-wire system with one conductor earthed.

For this system,

$$I_1 = \frac{P}{V_m}$$

Let the cross-sectional area of each conductor be a_1 .

$$R_1 = \frac{\rho l}{a_1}$$

Line loss:

$$P_{\text{loss}} = 2I_1^2 R_1 = 2 \left(\frac{P}{V_m} \right)^2 \frac{\rho l}{a_1}$$

Therefore,

$$a_1 = \frac{2P^2 \rho l}{P_{\text{loss}} V_m^2}$$

Volume of conductor material:

$$V_{dc} = 2a_1 l = \frac{4P^2 \rho l^2}{P_{\text{loss}} V_m^2}$$

Let,

$$K = \frac{4P^2 \rho l^2}{P_{\text{loss}} V_m^2}$$

So,

$$\boxed{V_{dc} = K}$$

Case 2: Two-phase three-wire overhead system

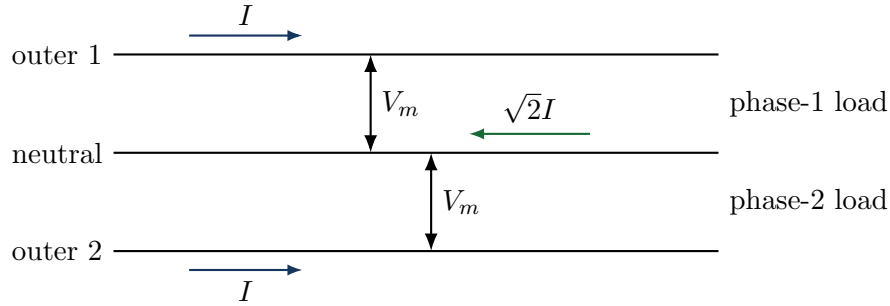


Figure 4: Two-phase three-wire system. The neutral carries the phasor sum of two currents.

For balanced two-phase load, power in each phase is $P/2$.

RMS voltage between outer and neutral is $V_m/\sqrt{2}$ in the standard overhead comparison basis. Hence,

$$\frac{P}{2} = \frac{V_m}{\sqrt{2}} I \cos \phi$$

$$I = \frac{P}{\sqrt{2} V_m \cos \phi}$$

The neutral current is the phasor sum of two phase currents which are 90° apart:

$$I_N = \sqrt{I^2 + I^2} = \sqrt{2}I$$

To keep current density same, area of neutral conductor is

$$a_N = \sqrt{2}a$$

Resistance of each outer conductor:

$$R = \frac{\rho l}{a}$$

Resistance of neutral conductor:

$$R_N = \frac{\rho l}{\sqrt{2}a} = \frac{R}{\sqrt{2}}$$

Total line loss:

$$P_{\text{loss}} = 2I^2 R + I_N^2 R_N$$

$$P_{\text{loss}} = 2I^2 R + (\sqrt{2}I)^2 \frac{R}{\sqrt{2}}$$

$$P_{\text{loss}} = I^2 R(2 + \sqrt{2})$$

Substituting I and R ,

$$P_{\text{loss}} = \left(\frac{P}{\sqrt{2}V_m \cos \phi} \right)^2 \frac{\rho l}{a} (2 + \sqrt{2})$$

$$a = \frac{P^2 \rho l (2 + \sqrt{2})}{2P_{\text{loss}} V_m^2 \cos^2 \phi}$$

Volume of conductor material:

$$V_{2\phi,3w} = 2al + \sqrt{2}al = a(2 + \sqrt{2})l$$

$$V_{2\phi,3w} = \frac{P^2 \rho l^2 (2 + \sqrt{2})^2}{2P_{\text{loss}} V_m^2 \cos^2 \phi}$$

Since,

$$K = \frac{4P^2 \rho l^2}{P_{\text{loss}} V_m^2}$$

$$V_{2\phi,3w} = \frac{(2 + \sqrt{2})^2}{8 \cos^2 \phi} K$$

Now,

$$\frac{(2 + \sqrt{2})^2}{8} = \frac{6 + 4\sqrt{2}}{8} \approx 1.457$$

Therefore,

$$V_{2\phi,3w} = \frac{1.457}{\cos^2 \phi} K$$

The two-phase three-wire overhead system requires $\frac{1.457}{\cos^2 \phi}$ times the conductor material required by the d.c. two-wire system with one conductor earthed.

4 Group A, Question 3: Kelvin's Law and Economic Conductor Size

Question 3

Mention the limitations of Kelvin's law. The daily load cycle of a three-phase transmission line, 20 km long, is equivalent to a current of 100 A for 8 hours, 75 A for 7 hours, and 20 A for 9 hours. The capital cost per km of line is Rs. $(9500 + 4000a)$, where a is the cross-sectional area in cm^2 of the conductor. Applying Kelvin's law, determine the most economical size of conductor. Interest and depreciation are 10% per annum, energy cost is 5 paise per unit, and resistance per km of conductor of 1 cm^2 cross-section is 0.19 ohm .

Limitations of Kelvin's law

1. It is difficult to estimate the actual annual load curve before installation.
2. The cost of supports, insulators and erection may not vary exactly with conductor area.
3. It does not consider mechanical strength requirement of the conductor.
4. It does not consider safe current density and heating limit properly.
5. It does not consider corona loss and skin effect in detail.
6. Interest and depreciation rates may change with time.
7. The economic size found may give excessive voltage drop, so voltage regulation must also be checked.

Kelvin's law statement

Kelvin's law

The most economical conductor size is obtained when the variable annual charge on the conductor is equal to the annual cost of energy wasted in the conductor.

$$\text{Variable annual capital charge} = \text{annual cost of energy loss}$$

Given data

Length of line = 20 km

Capital cost per km = Rs. $(9500 + 4000a)$

Interest and depreciation = 10%

Energy cost = 5 paise/kWh = Rs. 0.05/kWh

Resistance per km for $1 \text{ cm}^2 = 0.19 \text{ } \Omega$

Daily current cycle:

100 A for 8 h, 75 A for 7 h, 20 A for 9 h

Resistance of each conductor

For area $a \text{ cm}^2$,

$$R = \frac{0.19 \times 20}{a} = \frac{3.8}{a} \Omega$$

Daily energy lost

For three-phase line,

$$\text{Power loss} = 3I^2R$$

Daily energy loss:

$$E_d = \frac{3R}{1000} [(100)^2 \times 8 + (75)^2 \times 7 + (20)^2 \times 9]$$

First calculate the bracket:

$$(100)^2 \times 8 = 80000$$

$$(75)^2 \times 7 = 39375$$

$$(20)^2 \times 9 = 3600$$

$$\text{Total} = 122975$$

Therefore,

$$E_d = \frac{3 \times (3.8/a) \times 122975}{1000}$$

$$E_d = \frac{1401.915}{a} \text{ kWh/day}$$

Annual energy loss:

$$E_y = \frac{1401.915 \times 365}{a} = \frac{511699}{a} \text{ kWh/year}$$

Annual cost of energy lost:

$$C_e = 0.05 \times \frac{511699}{a} = \frac{25584.95}{a} \text{ rupees/year}$$

Annual variable capital charge

Variable part of capital cost per km:

$$4000a$$

For 20 km:

$$4000a \times 20 = 80000a$$

Annual charge at 10%:

$$C_c = 0.10 \times 80000a = 8000a$$

Apply Kelvin's law

$$C_c = C_e$$

$$8000a = \frac{25584.95}{a}$$

$$a^2 = \frac{25584.95}{8000} = 3.1981$$

$$a = \sqrt{3.1981} = 1.788 \text{ cm}^2$$

Most economical conductor area is approximately 1.79 cm^2 .

Exam note

After applying Kelvin's law, the selected conductor should also be checked for voltage drop, temperature rise, mechanical strength, and corona condition. Kelvin's law alone is not enough in practical design.

5 Group A, Question 4(i): Ring Main Distributor

Question 4(i)

A 2-wire d.c. distributor ABCDEA is in the form of a ring main and is fed at point A at 230 V. It is loaded as follows: 20 A at B, 40 A at C, 60 A at D, and 20 A at E. The resistance of various sections, including go and return, are: AB = 0.1 Ω , BC = 0.05 Ω , CD = 0.01 Ω , DE = 0.025 Ω , EA = 0.075 Ω . Determine the point of minimum potential and current in each section of the distributor.

Ring main idea

A ring main distributor forms a closed loop. For solving, assume current in one section and apply KVL around the closed ring:

$$\sum IR = 0$$

Given ring main diagram

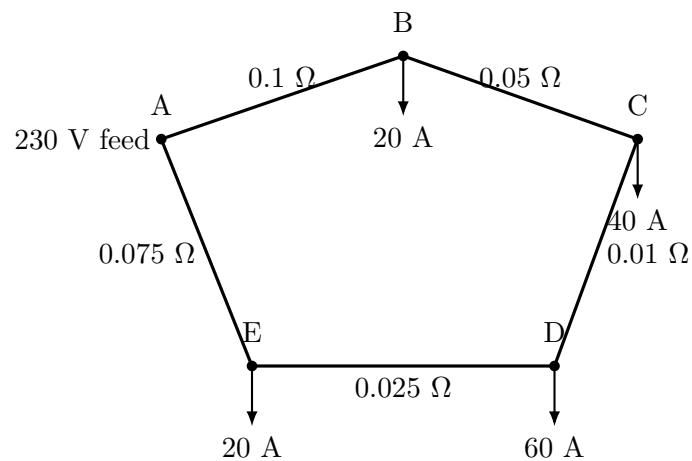


Figure 5: Ring main distributor ABCDEA.

Step 1: Assume current

Let current in section AB be I ampere in clockwise direction.

Then section currents are:

$$I_{AB} = I$$

$$I_{BC} = I - 20$$

$$I_{CD} = I - 20 - 40 = I - 60$$

$$I_{DE} = I - 20 - 40 - 60 = I - 120$$

$$I_{EA} = I - 20 - 40 - 60 - 20 = I - 140$$

Step 2: Apply KVL around the ring

$$I(0.1) + (I - 20)(0.05) + (I - 60)(0.01) + (I - 120)(0.025) + (I - 140)(0.075) = 0$$

Expanding,

$$0.26I - 15.1 = 0$$

$$I = \frac{15.1}{0.26} = 58.08 \text{ A}$$

Step 3: Find currents in all sections

Section	Algebraic current	Actual direction and value
AB	$I = 58.08$	58.08 A from A to B
BC	$I - 20 = 38.08$	38.08 A from B to C
CD	$I - 60 = -1.92$	1.92 A from D to C
DE	$I - 120 = -61.92$	61.92 A from E to D
EA	$I - 140 = -81.92$	81.92 A from A to E

Step 4: Point of minimum potential

Current flows towards point C from both sides:

- From A to B to C, current reaches C.
- From A to E to D to C, current also reaches C.

Therefore, the point of minimum potential is C.

Currents: AB = 58.08 A A to B; BC = 38.08 A B to C; CD = 1.92 A D to C; DE = 61.92 A E to D; EA = 81.92 A A to E. Minimum potential point is C.

6 Group A, Question 4(ii): Conductor Volume Reduces with Increase in Transmission Voltage

Question 4(ii)

Show that the volume of conductor material reduces with increase in transmission voltage.

Basic idea

For the same power, if transmission voltage increases, current decreases. Since line loss depends on $I^2 R$, the required conductor area becomes smaller. Hence conductor volume decreases.

Derivation

For a three-phase transmission line,

$$P = \sqrt{3} V I \cos \phi$$

So,

$$I = \frac{P}{\sqrt{3} V \cos \phi}$$

Line loss:

$$W = 3 I^2 R$$

For one conductor,

$$R = \frac{\rho l}{a}$$

Therefore,

$$W = 3 \left(\frac{P}{\sqrt{3} V \cos \phi} \right)^2 \frac{\rho l}{a}$$

$$W = \frac{P^2 \rho l}{V^2 \cos^2 \phi a}$$

So,

$$a = \frac{P^2 \rho l}{W V^2 \cos^2 \phi}$$

Volume of three conductors:

$$\text{Volume} = 3 a l$$

$$\text{Volume} = \frac{3 P^2 \rho l^2}{W V^2 \cos^2 \phi}$$

Thus,

$$\boxed{\text{Volume} \propto \frac{1}{V^2}}$$

As transmission voltage increases, required conductor volume decreases according to $\text{Volume} \propto 1/V^2$.

Simple flowchart

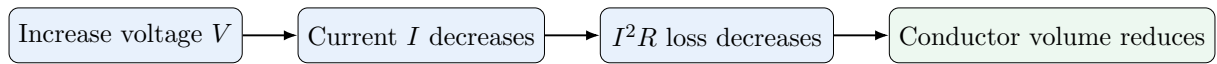


Figure 6: Effect of increasing transmission voltage.

7 Group A, Question 5: Radial Distribution and Distributor Numerical

Question 5

What are the drawbacks of radial distribution system? The resistance of each conductor of the 2-wire d.c. distributor shown in the figure is 0.02 ohm/km . Determine the point of minimum potential and the minimum voltage.

Drawbacks of radial distribution system

1. Supply reliability is poor because fault in one feeder interrupts supply to all consumers beyond the fault point.
2. Voltage regulation is poor at distant load points.
3. Consumers near the feeding point get better voltage than distant consumers.
4. It is not suitable for important urban or industrial loads where continuity is required.
5. Load sharing is not flexible.
6. Expansion of system may increase voltage drop heavily.

Given distributor diagram

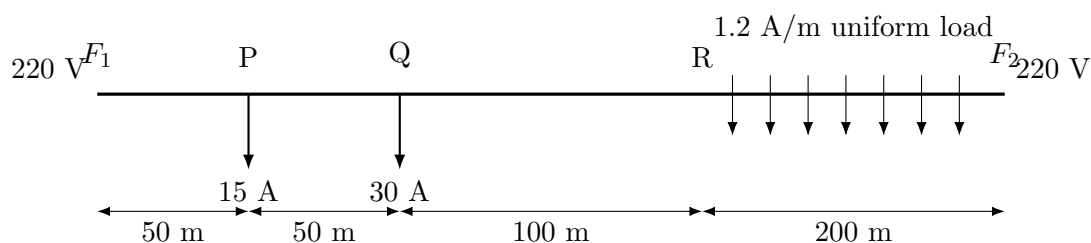


Figure 7: 2-wire d.c. distributor fed at both ends at equal voltage.

Given data

$$V_{F_1} = V_{F_2} = 220 \text{ V}$$

Resistance of each conductor:

$$0.02 \text{ } \Omega/\text{km}$$

Resistance of distributor including go and return:

$$r = 2 \times 0.02 = 0.04 \text{ } \Omega/\text{km} = 0.00004 \text{ } \Omega/\text{m}$$

Uniform load from R to F_2 :

$$1.2 \text{ A/m over } 200 \text{ m}$$

Total uniform load:

$$1.2 \times 200 = 240 \text{ A}$$

Step 1: Assume current from left end

Let current supplied from F_1 be I ampere.

Section currents:

$$F_1P : I$$

$$PQ : I - 15$$

$$QR : I - 15 - 30 = I - 45$$

At a distance y metre from R towards F_2 , current in the uniformly loaded section is

$$I - 45 - 1.2y$$

Step 2: Equal end voltages

Since both ends are at 220 V,

$$\text{total voltage drop from } F_1 \text{ to } F_2 = 0$$

Hence,

$$I(50)r + (I - 15)(50)r + (I - 45)(100)r + \int_0^{200} (I - 45 - 1.2y)r \, dy = 0$$

Putting $r = 0.00004$,

$$0.016I - 1.53 = 0$$

$$I = 95.625 \text{ A}$$

Step 3: Point of minimum potential

In the uniform-load section, minimum point occurs where current becomes zero:

$$I - 45 - 1.2y = 0$$

$$95.625 - 45 - 1.2y = 0$$

$$1.2y = 50.625$$

$$y = 42.1875 \text{ m}$$

Therefore, point of minimum potential is 42.19 m from R towards F_2 .

Distance from F_1 :

$$50 + 50 + 100 + 42.19 = 242.19 \text{ m}$$

Step 4: Minimum voltage

Voltage drop from F_1 to P:

$$= 95.625 \times 50 \times 0.00004 = 0.19125 \text{ V}$$

Voltage drop from P to Q:

$$= (95.625 - 15) \times 50 \times 0.00004 = 0.16125 \text{ V}$$

Voltage drop from Q to R:

$$= (95.625 - 45) \times 100 \times 0.00004 = 0.2025 \text{ V}$$

Voltage drop from R to minimum point:

$$\begin{aligned} &= \int_0^{42.1875} (50.625 - 1.2y) 0.00004 \, dy \\ &= 0.00004 \left[50.625y - 0.6y^2 \right]_0^{42.1875} \\ &= 0.04272 \text{ V} \end{aligned}$$

Total drop:

$$= 0.19125 + 0.16125 + 0.2025 + 0.04272 = 0.59772 \text{ V}$$

Minimum voltage:

$$V_{min} = 220 - 0.59772 = 219.402 \text{ V}$$

Point of minimum potential is 42.19 m from R towards F_2 , or
242.19 m from F_1 . Minimum voltage is 219.40 V.

8 Group B, Question 6(i): Functions of Instrument Transformers in a Substation

Question 6(i)

Explain the functions of instrument transformers in a substation.

Instrument transformers

Instrument transformers are transformers used with measuring instruments and protective relays. They reduce high current or high voltage to a safe, measurable value.

There are two main types:

- Current Transformer (CT)
- Potential Transformer (PT) or Voltage Transformer (VT)

Simple diagram

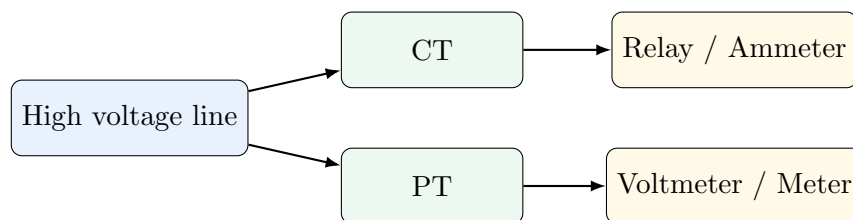


Figure 8: Use of CT and PT in a substation.

Functions

1. They reduce high current and high voltage to safe values.
2. They isolate measuring instruments and relays from high-voltage circuits.
3. They allow standard instruments to be used in high-voltage substations.
4. CT supplies proportional current to ammeters, wattmeters and relays.
5. PT supplies proportional voltage to voltmeters, energy meters and relays.
6. They help in protection by feeding current and voltage signals to relays.
7. They improve safety of operating staff and measuring equipment.

Typical standard secondary values

Transformer	Primary quantity	Secondary quantity
CT	High current	5 A or 1 A
PT	High voltage	110 V or 100 V

Instrument transformers make measurement and protection safe, economical and standardised in a substation.

9 Group B, Question 6(ii): HRC Fuse and Circuit Breaker Advantages

Question 6(ii)

Explain the principle of operation of HRC fuse. What are the advantages of a circuit breaker over a fuse?

HRC fuse

HRC means High Rupturing Capacity. An HRC fuse can safely interrupt very high fault current without explosion or fire.

Neat labelled diagram of HRC fuse

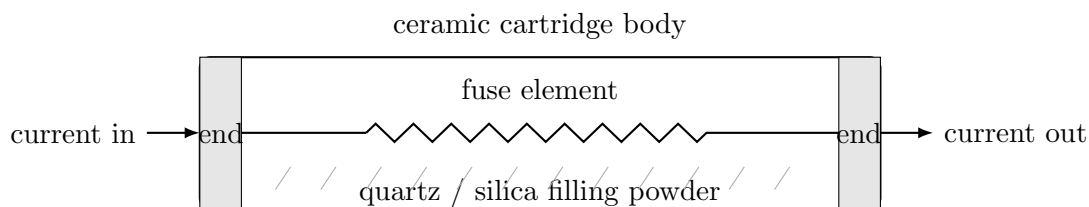


Figure 9: Construction of HRC cartridge fuse.

Principle of operation

1. Under normal condition, the fuse element carries rated current safely.
2. During fault, current increases suddenly.
3. Heat produced in the fuse element is proportional to I^2R .
4. Due to high fault current, the fuse element melts quickly.
5. An arc is formed after melting.
6. The filling powder absorbs heat and helps to quench the arc.
7. Thus the faulty circuit is opened safely.

Fuse operation flowchart

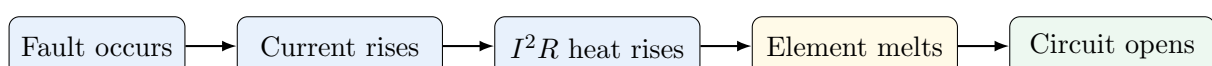


Figure 10: Operation sequence of HRC fuse.

Advantages of circuit breaker over fuse

Fuse	Circuit breaker
Works only once and must be replaced after operation.	Can be reset and reused after operation.
Cannot be used conveniently for remote control.	Can be operated manually or automatically from remote place.
Protection characteristic is fixed.	Relay settings can be adjusted.
Not suitable for frequent switching.	Suitable for frequent switching and protection.
Only interrupts fault by melting.	Interrupts fault mechanically using trip mechanism.

HRC fuse protects by melting of fuse element and arc quenching by filling powder. Circuit breaker is superior because it is reusable, controllable, adjustable and suitable for repeated operation.

10 Group B, Question 7(i): Objective of Earthing and Plate Earthing

Question 7(i)

What is the primary objective of earthing electrical equipment? Discuss plate earthing.

Earthing

Earthing means connecting the metallic body of electrical equipment to the general mass of earth through a low-resistance conductor.

Primary objective of earthing

The main objective of earthing is to protect human life and equipment.

- It provides a low-resistance path for fault current.
- It keeps the metallic body of equipment nearly at earth potential.
- It prevents electric shock to a person touching the equipment.
- It helps protective devices like fuse or circuit breaker to operate quickly.
- It protects equipment from leakage current and overvoltage.

Plate earthing diagram

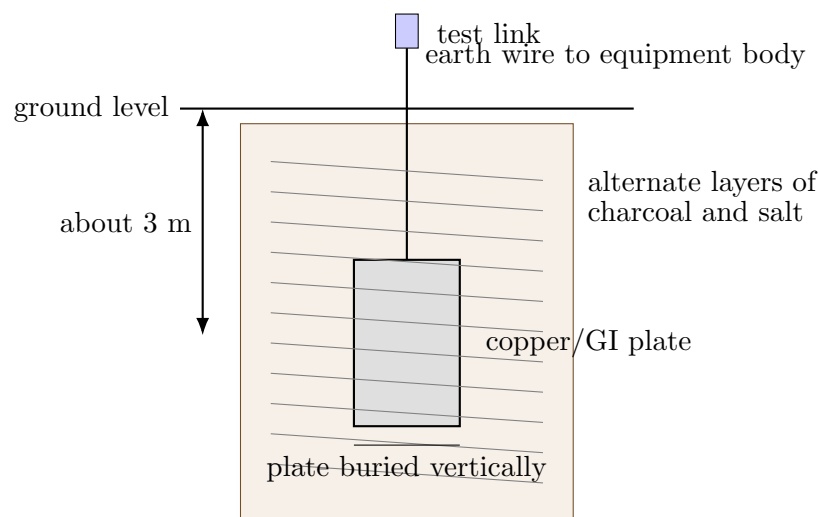


Figure 11: Plate earthing arrangement.

Construction and working

1. A copper or galvanized iron plate is buried vertically in the earth.
2. The plate is surrounded by alternate layers of charcoal and salt.

3. Charcoal and salt reduce earth resistance by retaining moisture.
4. The earth wire is bolted to the plate with proper nut and washer.
5. A watering pipe may be provided to keep the soil moist.
6. During fault, current flows safely to earth through the earth wire and plate.

Common sizes

Plate material	Common size
Copper plate	60 cm × 60 cm × 3.18 mm
GI plate	60 cm × 60 cm × 6.35 mm

Earthing protects people and equipment by providing a low-resistance path to earth. In plate earthing, a copper or GI plate is buried vertically and connected to the equipment body.

11 Group B, Question 7(ii): Compact Fluorescent Lamp

Question 7(ii)

Explain the principle of operation of compact fluorescent lamp.

Compact fluorescent lamp

A compact fluorescent lamp, or CFL, is a small fluorescent lamp. It produces light by electric discharge through mercury vapour and conversion of ultraviolet radiation into visible light by phosphor coating.

Labelled diagram

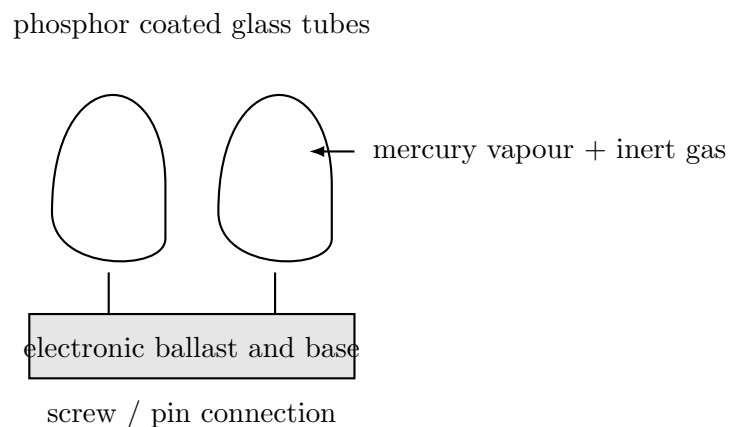


Figure 12: Basic construction of compact fluorescent lamp.

Principle of operation

1. When supply is switched on, the electronic ballast gives proper starting voltage.
2. Current flows through electrodes and heats them.
3. Electrons are emitted from the heated electrodes.
4. These electrons collide with mercury vapour atoms inside the tube.
5. Mercury atoms emit ultraviolet radiation.
6. The phosphor coating on the tube absorbs ultraviolet radiation.
7. The phosphor coating emits visible light.

Operation flowchart

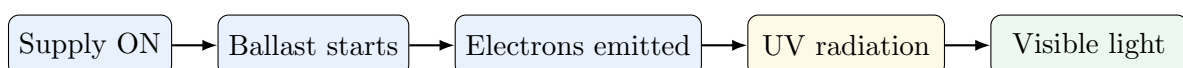


Figure 13: Working sequence of CFL.

Advantages

- It consumes less energy than incandescent lamp.
- It produces less heat.
- It has longer life.
- It gives better luminous efficiency.

CFL works by gas discharge in mercury vapour. UV radiation produced inside the tube is converted into visible light by phosphor coating.

12 Group B, Question 8(i): Advantages of High Tension Service Main Connection

Question 8(i)

What are the advantages of high tension service main connections?

High tension service main

High tension service main connection means supply is given to the consumer at high voltage, usually for large load consumers. The consumer generally has a transformer substation in the premises.

Advantages

1. It gives tariff advantage for large consumers.
2. It reduces current for the same load, so voltage drop is reduced.
3. It gives better voltage regulation.
4. It reduces interference with supply of neighbouring consumers.
5. It permits use of large motors, welding plants and heavy industrial loads.
6. It gives better earth-fault protection for heavy current circuits.
7. The consumer gets better control of supply through own substation.

Simple layout



Figure 14: High tension service main connection.

High tension service main connection is suitable for large consumers because it gives better voltage regulation, lower current, better protection and tariff advantage.

13 Group B, Question 8(ii): General Stipulations for Low Tension Service Main Connection

Question 8(ii)

Mention the general stipulations to be followed for low tension service main connection.

Low tension service main

Low tension service main is the connection from the supply company's service line to the consumer's main switch and distribution board at low voltage.

General stipulations

1. Power and heating sub-circuits should be kept separate from lighting and fan sub-circuits.
2. Wiring should be done through main distribution board and sub-distribution boards located near load centres.
3. Conductors should be of suitable size and current rating.
4. Voltage drop from consumer terminal to any point should not exceed the prescribed limit, usually about 3%.
5. Fuse should be provided on the live conductor, not on neutral.
6. Neutral wire should not be broken by switch or fuse.
7. All switches should be connected in the live wire.
8. All metallic switch boards and distribution boards should be properly earthed.
9. Socket outlets should preferably have three terminals: live, neutral and earth.
10. Lights and fans may be connected in one sub-circuit, but one sub-circuit should not have more than 10 points or 800 W load.
11. Each branch circuit should be protected by a suitable fuse or miniature circuit breaker.
12. Load should be distributed as evenly as possible among different circuits.

Low tension service main flow diagram

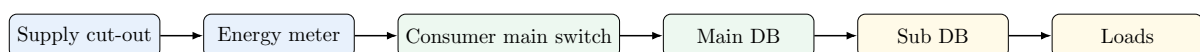


Figure 15: Low tension service main connection sequence.

SN ma'am exam remark

In low tension wiring answers, always mention: separate circuits, fuse on live wire, neutral not broken, earthing of metallic parts, voltage drop limit, and maximum 10 points or 800 W per lighting/fan sub-circuit.

Low tension service main connection must be safe, properly protected, properly earthed, and arranged through suitable distribution boards with live-wire switching and correct fuse protection.

Final Revision Table

Question	Core idea	Final result / memory point
Q1(i)	Unequal-voltage distributor	$x = \frac{V_A - V_B}{ir l} + \frac{l}{2}$
Q1(ii)	D.C. to 3-phase conversion	Additional load = 62% at 0.9 p.f.
Q2	Conductor comparison	Two-phase 3-wire needs $\frac{1.457}{\cos^2 \phi} K$
Q3	Kelvin's law	Economic area $\approx 1.79 \text{ cm}^2$
Q4(i)	Ring main distributor	Minimum potential at C
Q4(ii)	Voltage and conductor volume	Volume $\propto 1/V^2$
Q5	Fed-both-end distributor	Minimum point: 42.19 m from R, $V_{min} = 219.40 \text{ V}$
Q6	CT/PT, HRC fuse	Safe measurement and safe fault interruption
Q7	Earthing, CFL	Low resistance path; UV to visible light
Q8	Service mains	HT advantages and LT safety stipulations

End of Part-II 2023 Complete Solution Set